

Introduction To Statistical Mechanics

4. Calculation of Mean Values for the Random Walk Problem

$$(a) W(n_1) = \frac{N!}{n_1!(N-n_1)!} p^{n_1} q^{N-n_1}$$

$$(b) \bar{n}_1 = Np$$

$$(c) \bar{m} = \overline{n_1 - n_2} = \bar{n}_1 - \bar{n}_2 = N(p - q)$$

$$(d) \overline{(\Delta n_1)^2} = Npq \quad (\text{Dispersion: Variance, Standard Deviation, Spread})$$

$$(e) \overline{(\Delta m)^2} = 4\overline{(\Delta n_1)^2} = 4Npq$$

Example: Consider the case of $N = 100$ steps, where $p = q = \frac{1}{2}$.

(a) Then the mean number $\bar{n}_1$ of steps to the right (or to the left) is 50;

(b) The mean displacement $\bar{m} = 0$.

(c) The root-mean square displacement $\left[\overline{(\Delta m)^2} \right]^{1/2} = 10$ steps.

Solve and check the results using previous example. Use little intuition.